GOKHALE MEMORIAL GIRLS' SCHOOL

PRE BOARD EXAMINATION - II (2025-26)

CLASS - X

SCIENCE

Time - 3 Hrs. **Full Marks - 80**

General Instructions :

i. This question paper consists of **39 questions** in 3 Sections. **Section - A is Biology, Section - B is Chemistry, Section - C is Physics**.

ii. All questions are compulsory. However an internal choice is provided in some questions. A student is expected to attempt only one of those questions.

Section - A (Biology)

1. Among the followings, an artificial ecosystem is — [1]

a) pond b) lake

c) grass land d) crop land

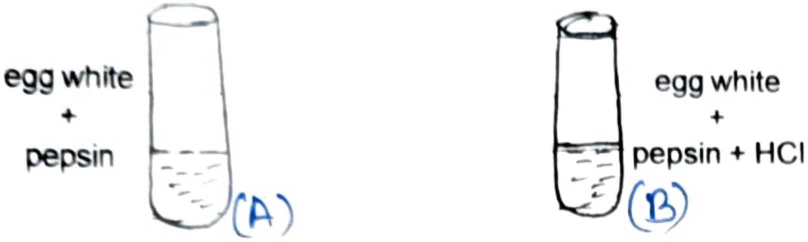
2. Opening and closing of stomatal pore depends upon — [1]

a) oxygen b) temperature

c) water in guard cells

d) concentration of O_2 in stomata

3. Study the following experimental setups & observe the role of enzymes in digestion of food. [1]



In which test tube the digestion of protein will occur? —

a) Test tube A as pepsin will breakdown protein into simple molecules.

b) Test tube B as HCl will breakdown protein into simple molecules

c) Test tube A as pepsin will get broken down into simple molecules

d) Test tube B as HCl will activate pepsin for breaking down protein into simpler molecules.

4. A cross between two tall pea plants resulted in offsprings having a few dwarf plants. The genotypes of the parental plants must be— [1]

a) Tt and Tt b) Tt and tt

c) TT and tt d) Tt and TT

5. In which of the following organisms food is broken and absorbed outside the body? [1]

a) Lice
b) Mushroom
c) Paramoecium
d) Amoeba

6. Select the row which has incorrect information. [1]

a) RrYy	Round - yellow
b) rrYY	Round - yellow
c) RrYY	Round - yellow
d) RRYY	Round - yellow

7. In a synapse, chemical signal is transmitted from — [1]

a) dendritic end of one neuron to axonal end of another neuron
b) axon to the cell body of the same neuron
c) cell body to axonal end of same neuron
d) axonal end of one neuron to dendritic end of another neuron

The following two questions consists of two statements — Assertion [A] and Reasons [R].

Answer these questions by selecting the appropiate option given below :

A) Both A and R are true, and R is the correct explanation of A.

B) Both A and R are true and R is not the correct explanation of A.

C) A is true but R is false.

D) A is false but R is true.

8. Assertion (A) – Free ear lobe is a dominant trait.

   Reasom (R) – It is more common as it is expressed both in homozygous & heterozygous condition. [1]

9. Assertion (A) – Auxin hormone that helps the cells to grow longer, is produced in the shoot tip.

   Reason (R) – Auxin tends to diffuse towards the shady side of the shoot. [1]

10. Name the hormone which — $[{}^{1}/_{2} \times 4 = 2]$

    i) is associated with dwarfism

    ii) maintains balance for growth

    iii) responsible for puberty changes in females

    iv) makes heart beat faster during scary situation

11. a) How and where in the kidneys is urea removed from blood?

OR

b) How is the suction force generated in xylem and what is the significance of this force? [2]

12. Who among human male and human female is considered as heterogametic and why? [2]

13. a) Construct a food web (as a word diagram) comprising ~~comparising~~ of the following organisms — grass, crop - plants, frog, field mouse, snake, grass-hopper, hawk.

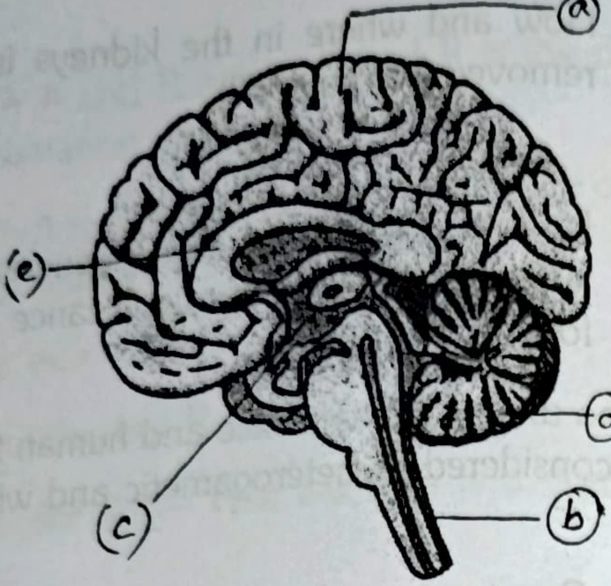
b) Define biological magnification. [2+1=3]

14. a) Why is the testis located outside the abdominal cavity?

b) State the functions of vas deferens and seminal ~~seninal~~ vesicles. [1+2=3]

15. Mala was attracted by a freshly blooming rose and went near it to smell it. As she was going to touch it fascinated by its smell, she touched a thorn and quickely moved her hands away.

Answer the following questions based on the following diagram and Mala's experience.



[Attempt subpart A or B]

A) Define reflex action, and give an example. [1]

OR

B) Which of Mals's reactions best demonstrates voluntary action and why? [1]

C) In the diagram, identify and name the part that controls reflex action & the part that controls the endocrine system. [1]

D) Write two difference between (a) and (d). [2]

16. A) Name the plant Mendel used for his experiment. Work out with the help of a cross, the type of progenies obtained by Mendel in

F_1 & F_2 generations, when he crossed tall and short plants. Write the phenotypic and genotypic ratios he obtained in F_2 generation plants. [1+2+2=5]

OR

B) Draw a neat & labelled (any four labellings) diagram of human heart and show the flow of deoxygenated blood only, through arrows.

Write one difference in composition and one difference in function of blood & lymph. [3+2=5]

Section - B (Chemistry)

17. The sum of all the coefficients of the different chemical species of the given equation is —

$NH_3 (g) + O_2 (g) \rightarrow NO (g) + H_2O (l)$

a) 24 b) 19

c) 17 d) 15 [1]

18. A hydrocarbon having two C – C single bond and one C ≡ C triple bond will be

a) Ethyne b) Propyne

c) Butyne d) Benzene [1]

19. In a metal - displacement reaction, if metal A can displace metal B from a salt of B, the reducing agent is

a) B   b) A   c) B^{+2} d) A^{+2} [1]

20. Which of the following correctly represents Sodium Oxide?

a) $Na^{+2} \cdot 2\left[\begin{array}{c} xx \\ xx\,O\,xx \\ xx \end{array}\right]^{2-}$

b) $2\left[Na^{+}\right]\left[\begin{array}{c} xx \\ xx\,O\,xx \\ xx \end{array}\right]^{2-}$

c) $2\left[Na^{+2}\right] 2\left[\begin{array}{c} xx \\ xx\,O\,xx \\ xx \end{array}\right]^{-}$

d) $Na^{+}\left[\begin{array}{c} xx \\ xx\,O\,xx \\ xx \end{array}\right]^{2-}$ [1]

21. Soaps do not form lather in hard - water because — [1]

a) Soaps are insoluble in water

b) Soaps react with Sodium - ions (Na^{+}) to form insoluble salts

c) Hard - water contains Ca^{+2} and Mg^{+2} - ions that form insoluble scum with soap

d) Soap molecules lose their cleansing property in hot - water.

22. Which of the following statement is/are incorrect when HCl (g) is passed through water – [1]

a) It does not ionise in solution as it is a covalent compound.

b) It gives both Hydrogen ion and Chloride ions in solution.

c) It forms Hydronium ions in solution due to the combination of Hydrogen - ion with water molecules.

d) The resultant solution will not change the colour of phenolphthalein indicator to pink.

23. The gas produced when metallic sodium is added to alcohol – [1]

a) turns limewater milky

b) makes a pop-sound when the burning splinter is introduced in it

c) turns white coloured anhydrous Copper Sulphate blue

d) has a pungent odour

The following question has two statements — Assertion (A) and Reason (R). Answer these question by selecting the appropriate option given below :

a) Both A & R are true, and R is the correct explanation of A.

b) Both A & R are true, but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

24. Assertion (A) : Silver Bromide decomposition is used in black and white photography.

    Reason (R) : Light provides the energy for the exothermic reaction. [1]

25. Give reason why — [1+1=2]

    a) The reaction of Iron (III) Oxide with Aluminium is used to join cracked machine parts (Provide the balanced chemical equation also)

    b) Sodium cannot be obtained from its ore by reduction with carbon.

26. X & Y are two elements, with atomic number 20 and 17, respectively.

a) Show the formation of the compound between X and Y, by electron transfer method.

b) State what would happen if electric current is passed through the aqueous solution of this compound? [2+1=3]

OR

26. a) Show the formation of Sodium Sulphide by electron transfer method (Atomic no. of S = [redacted], Na = [redacted]]
16 11

b) Name the alloy and mention the constituting elements of the alloy, that has a low melting point and used for welding of electrical wires. [2+1=3]

27. Explain what happens and give suitable chemical reaction, when —

a) Excess Carbon di-oxide gas is passed through Lime water.

b) A piece of metallic Zinc is dropped in blue Copper Sulphate solution.

c) Potassium Iodide solution is added to Lead Nitrate Solution. [1+1+1=3]

28. A) a) 'X' forms the major constituent of the tooth enamel. It is the hardest substance in the human body which

is insoluble in water, but undergoes corrosion if the pH value decreases below 5.5. In the mouth, the bacteria present produces mild acid due to degradation of sugar and fodd particle, which results in tooth decay.

i) What is the chemical name and molecular formula of 'X'?

ii) Mention how this decay can be prevented and why?

b) i) What noticeable changes would you observe when Copper Sulphate crystals are heated in a clean, dry boiling tube?

ii) What happens when 2-3 drops of water is added to the sample in the tube?

Explain the phenomenon that occurs in b (i) & b (ii) with suitable equations. [(1+1) + (1+1) = 4]

OR

28. B) Rajesh was given a substance, 'A' and was asked to identify it. He conducted three tests — Test I, II and III on 'A' and recorded the following results.

Test I – On heating, the sample releases Carbon-di-oxide, water and a Sodium Salt.

Test II – The sample turns universal indicator to greenish-blue clour

Test III– He could prepare the sample, using ammonia as a raw material.

a) Identify the substance 'A' that was given to Rajesh.

b) Mention an use of 'A', based on the properties observed in Test II.

c) Rajesh observed that recrystallisation of the sodium salt formed in Test I, gives a basic salt which is used in the manufacture of Borax. Identify that basic salt.

d) Write the chemical reaction that occurs in Test III. [1+1+1+1=4]

29. A) The Carbon Compound, 'A' on heating with excess of concentrated H_2SO_4 forms the compound, 'B'. 'B' produces the compound 'C' on treating with one mole of Hydrogen gas in presence of Nickel - catalyst. On combustion in air, 'C' gives 2 moles of CO_2 and 3-moles of H_2O.

a) Identify A, B and C

b) Give all the chemical equations involved.

c) State the role of concentrated Sulphuric Acid in the formation of B from A. [3+1.5+0.5=5]

OR

29. 'P' is an organic compound of molecular formula $C_2H_4O_2$, which can be used as a food - preservative. 'P' reacts with 'Q' (C_2H_6O) in presence of an inorganic acid to give 'R' that has a sweet - smell.

a) Write the IUPAC - name and the molecular structure of 'P'.

b) Identify 'Q' and 'R'.

c) What is the role of the inorganic acid used in the formation of R from P and Q.

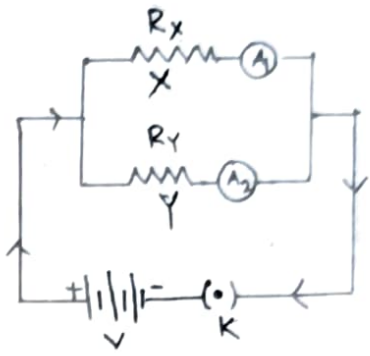
d) Give all the chemical equations involved in the reactions that take place.

e) What happens when 'P' reacts with NaOH? Name the product formed.

[1+2+0.5+1+0.5=5]

Section - C (Physics)

30. In the circuit diagram shown, the two resistance wires X and Y are of same length and same material but X is more thicker than Y. Ammeter A_1 and A_2 are connected in the ciruit. Choose from the following, the correct option.



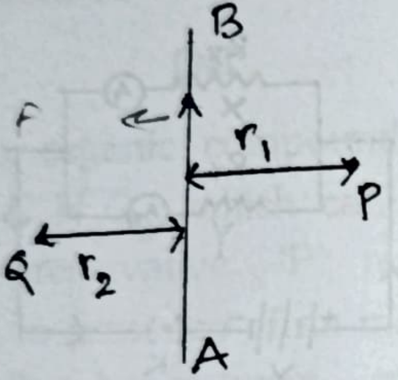
a) Resisitance of X = Resistance of Y and reading of ammeter A_1 = reading of ammeter A_2.

b) Resistance of X > Resistance of Y and reading of ammeter A_1 > Reading of ammeter A_2.

c) Resistance of X < Resistance of Y and reading of ammeter A_1 > Reading of ammeter A_2.

d) Resistance of X < Resistance of Y and reading of ammeter A_1 < Reading of ammeter A_2. [1]

31. An infinite straight current carrying conductor is placed perpendicular to the plave of paper. Current flows from A to B. Two points P and Q are marked at a distance r_1 and r_2 on the right and left of the conductor in the plane of paper such that $r_1 > r_2$.



I. Direction of magnetic field at point P is opposite to the direction of magnetic field at point Q.

II. Strength of magnetic field at point Q is greater than the strength of magnetic field at P.

III. If the direction of current is reversed, direction of magnetic field at P is opposite to the direction of magnetic field at Q and strength of magnetic field at point Q is less than the strength of magnetic field at point P. [1]

Choose from the following, the correct option that lists the correct statements about the magnetic field.

a) I and II

b) I, II and III

c) II and III

d) I and III

The following questions consists of two statements — Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below.

a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

c) (A) is true, but (R) is false.

d) (A) is false, but (R) is true.

32. Assertion (A) : A light ray incident along the line joining the center of curvature to any point on the spherical concave mirror will reflect along the same path.

Reason (R) : Incident ray strikes the mirror at a right angle making angle of incidence equal to angle of reflection. [1]

33. A person needs a lens of power –2.5 D for correcting his vision.

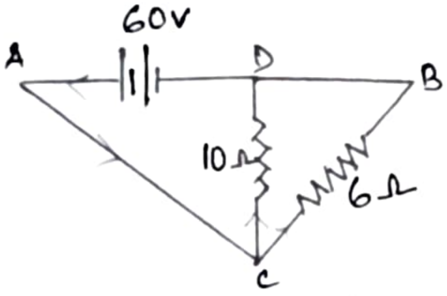
a) Name the defect of vision he is suffering from? Identify the lens.

b) Where is the far point of this myopic eye.

c) What is the actual distance of far point of the myopic eye? [2]

34. Attempt either option A or B.

A) Find out the following in the electric circuit given in the figure.



a) Effective resistance of 10Ω and 6Ω.

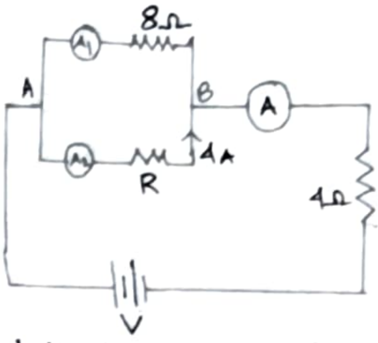
b) Current flowing through the circuit and the potential difference across 6Ω resistor. [2]

OR

B) Study the circuit, if voltage across AB = 16V, reading of ammeter A_2 and A are respectively 4A and 6A; Find

a) Voltage across 4Ω resistor, and reading of ammeter ~~A_2~~ A_1

b) Find R, and calculate equivalent resistance. [2]



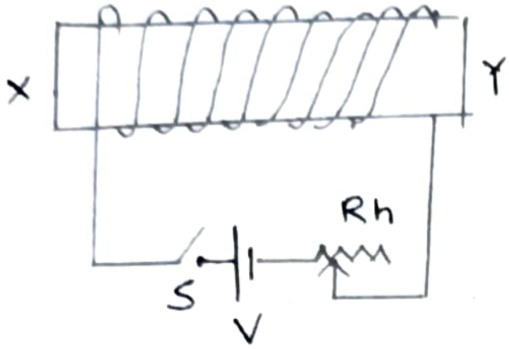
35. An object and its image are shown in the diagram for a spherical lens;

i) If the distance between object and image is 4 cm and the magnification is 3, find, distance of object, image from the optical centre, find its focal length. Draw the ray diagram showing the position of lens object pin and image pin. [3]

36. a) A 5Ω resistor is connected in series with a parallel combination of n-resistors of 6Ω each. The equivalent resistance is 7Ω . Find n.

b) A uniform wire of resistance 7Ω has a length of 2m. Find the length of the similar wire which connected in parallel with 2m long wire of same cross-section, will give a resistance of 5Ω. [3]

37. Spiral coil wound around a core XY. Current is switched on by closing the key.

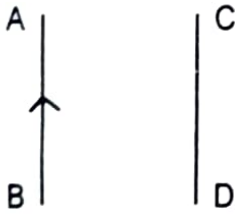


a) What will be the polarity at the ends X and Y.

b) How will the compass needle be affected placed near the end of X. Justify.

c) How can we increase the strength of an electro - magnet? [3]

38. A student performed an experiment to check the motion of a conductor (straight conductor) placed in the magnetic field of the current carrying conductor AB, carrying 10A current.



a) If a conductor CD is placed as shown, what is the direction of magnetic field at CD due to current carrying conduct AB. Now current is allowed to flow through CD from D to C; in which direction conductor carrying current CD gets deflected.

b) Whether an alpha particle will experience any force in a magnetic field if (i) it is placed in the field rest (ii) if it moves in the magnetic field perpendicular to the field lines.

Justify by stating the relevant principle/ rule. [5]

39. Attempt either option A or B.

a) The arrangement of two bulbs A and B are rated 40W, 240V and 60W, 240V are connected in series. With a electric main supply of 240V.

i) Calculate the resistance of each lamp.

ii) Find the current drawn by the two bulbs from the line.

iii) Find power dissipation across each bulb.

iv) What is the total energy consumed per unit second by the two bulbs.

[4]

OR

B) Two resistors A and B of resistance 4Ω and 6Ω respectively are connected in parallel. The combination is connected across a 6V battery of negligible resistance. Calculate

i) Power supplied by the battery.

ii) Power dissipated in each resistor.

iii) If the supply voltage falls to 3V, how is the power consumed by the ~~lamp~~ resistors affected.

A fuse is rated 8A. Can it be used with an electrical appliance of rating 5kw, 200V. Justify. [4]

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